SelectBoost4Beta: Improving Variable Selection in Beta Regression Models

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Conclusion

Extension of selectBoost (Bertrand et al., 2021) algorithm to Beta regression models.

The selectBoost algorithm is a new general algorithm known to improve the precision of any existing variable selection method.

This algorithm

- is based on highly intensive simulations and
- takes into account the correlation structure of the data.

It can either produce

- a confidence index for variable selection or
- be used to design experiments.

Method

Overcome Linear Correlation

The selectBoost algorithm has been designed in a general framework to avoid to select non-predictive correlated features. The main goal is to improve the PPV, i.e. the proportion of selected variables which truly belong to the model.

Correlated Resampling

The main idea of our algorithm is to consider that any observed value of a group of linearly correlated variables of the $\mathbf{X}$ matrix is the independent realization of a given random function, which is then used to perturb the observed values of the relevant correlated variables.

Spherical Distribution

As we assume that the variables are centered and that $\|\mathbf{x}_{p\cdot}\|^2 = 1$ for $p = 1, ..., P$, we know that $\mathbf{x}_{p\cdot} \in \mathcal{S}^{N-2}$. As a consequence, we use a spherical distribution for resampling. For instance, a multivariate Gaussian distribution assumption for $\mathbf{X} = (\mathbf{x}_{1\cdot}, ..., \mathbf{x}_{P\cdot})$ advocates for the use a von Mises-Fisher distribution (Sra, 2012).

Availability

Soon on available on github:

`https://fbertran.github.io/selectboost4beta/`

Setting and Aim

Variable Selection is a Commonly Required Step

Technological innovations make it possible to measure large amounts of data in a single observation. As a consequence, problems in which the number P of variables is larger than the number N of observations have become common.

Penalized Likelihood

There is a vast literature dealing with the problem of variable selection in both statistical and machine learning areas (Fan and Lv, 2010). The main variable selection methods can be gathered in the common framework of penalized likelihood. The estimate $\hat{\boldsymbol{\beta}}$ is then given by:

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}\in\mathbb{R}^P}{\arg\min}\left[-\ell_N(\boldsymbol{\beta}) + \sum_{p=1}^{P} \mathrm{pen}_\lambda(\beta_p)\right]$$

where $\ell_N(.)$ is the log-likelihood function, $\mathrm{pen}_\lambda(.)$ is a penalty function and $\lambda \in \mathbb{R}$ is the regularization parameter. As the goal is to obtain a sparse estimation of the vector of parameters $\boldsymbol{\beta}$, a natural choice for the penalty function is to use the so-called $\mathcal{L}_0$ norm ($\|.\|_0$) which corresponds to the number of non-vanishing elements of a vector:

$$\begin{array}{rcl} \mathrm{pen}_\lambda : \mathbb{R} & \mapsto & \{0, \lambda\} \\ x & \mapsto & \begin{cases} \mathrm{pen}_\lambda(x) = \lambda & \text{if } x \neq 0 \\ \mathrm{pen}_\lambda(x) = 0 & \text{else} \end{cases} \end{array}$$

which induces $\sum_{p=1}^{P} \mathrm{pen}_\lambda(\beta_p) = \lambda \|\boldsymbol{\beta}\|_0$.

Beta Distribution

Let Γ be the Euler gamma function. When it is non-zero, the density function of the $\mathrm{Beta}(p, q)$ distribution is given by:

$$\pi(y; p; q) = \frac{\Gamma(p+q)}{\Gamma(p)\Gamma(q)} y^{p-1}(1-y)^{q-1}, \quad 0 < y < 1 \tag{1}$$

with $p > 0, q > 0$. The expectation and the variance of Y, a $\mathrm{Beta}(p, q)$ distributed variable, are equal to:

$$\mathbb{E}(Y) = \frac{p}{p+q} \quad \text{et} \quad \mathrm{Var}(Y) = \frac{pq}{(p+q)^2(p+q+1)}. \tag{2}$$

In order to be able to apply techniques similar to those used for generalised linear models by McCullagh and Nelder (1995), Ferrari et al. (2004) reparametrise the $\mathrm{Beta}(p, q)$ distribution as follows: $\mu = p/(p+q)$ and $\phi = p+q$, i.e. $p = \mu\phi$ and $q = (1-\mu)\phi$. Eq (2) then becomes:

$$\mathbb{E}(Y) = \mu \quad \text{et} \quad \mathrm{Var}(Y) = \frac{\mu(1-\mu)}{1+\phi}.$$

Thus μ is the mean value of the response and ϕ can be interpreted as a precision parameter. With these new parameters, the density function given in Eq (1) is equal to:

$$\pi(y; \mu; \phi) = \frac{\Gamma(\phi) y^{\mu\phi-1}(1-y)^{(1-\mu)\phi-1}}{\Gamma(\mu\phi)\Gamma((1-\mu)\phi)}, \quad 0 < y < 1. \tag{3}$$

Beta Regression: the Model

The practical value of the Beta distribution has been asserted several times, for example by Johnson et al. (1995): "Beta distributions are very versatile and a variety of uncertanties can be usefully modelled by them. This flexibility encourages its empirical use in a wide range of applications." Several recent articles have focused on the study of Beta regression and its properties, in particular Ferrari et al. (2004) for an introduction to these models and Kosmidis and Firth (2010), Simas et al. (2010) and Grün et al. (2012) for extensions or improvements to the estimation techniques of these models. The Beta regression model can of course be used as soon as the response Y has values in a bounded interval $[a; b]$, with $a < b$ fixed and known.

It is not possible to make direct use of the theory of the generalised linear model (Nelder et al., 1972), as this is based on the use of a family of laws put into natural exponential form (NEF). However, as Morris (1982) pointed out, the family of Beta laws, with the parameterisation adopted by Ferrari et al. (2004) and introduced above, is certainly a univariate exponential family, but not a natural exponential family. The latter are characterised by their variance function.

Let $Y_1, \ldots, Y_n$ be independent random variables distributed according to the density function given in Eq (3) with mean μ_t and unknown precision ϕ. We obtain the Beta regression model by assuming that the mean of $Y_t, 1 \leq t \leq n$, can be written as:

$$g(\mu_t) = \sum_{i=1}^{k} x_{ti}\beta_i = \eta_k$$

where $\beta = (\beta_1, \ldots, \beta_k)'$ is a vector of unknown regression parameters ($\beta \in \mathbb{R}^k$), $'$ the vector or matrix transpose, and $x_{t_1}, \ldots, x_{t_k}$ are the observed values of the k predictors, with $k < n$, and that are supposed known and fixed. Finally, $g(\cdot)$ is a strictly monotoneous real valued link function, two times differentiable, surjective and defined on the interval $]0; 1[$. The variance of Y_t is a function of μ_t and hence a function of the covariables. As a consequence, the model automatically accounts for some lacks of homoscedasticity.

There are several usual choices for the link function $g(\cdot)$. For example, the logit link $g(\mu) = \log(\mu/(1-\mu))$, the probit link $g(\mu) = \Phi^{-1}(\mu)$, with Φ the distribution function of the centred and reduced normal distribution, the complementary log-log link $g(\mu) = \log(-\log(1-\mu))$, the log-log link $g(\mu) = -\log(-\log(\mu))$. The use of the logit link makes it possible to interpret the exponential of the coefficients of the covariates in terms of the odds ratio.

The SelectBoost Algorithm

1. To use the SB algorithm, we need a grouping method gr_{c_0} depending on a user-provided constant $0 \leq c_0 \leq 1$. It must satisfy:
   $$select \quad : \quad \mathbb{R}^{N\times P} \times \mathbb{R}^N \quad \rightarrow \quad \{0, 1\}^P$$
   Let $gr_{c_0}(p)$ be the set of all variables which are considered to be linked to the variable $\mathbf{x}_p$. $\mathbf{X}_{gr_{c_0}(p)}$ is the submatrix of $\mathbf{X}$ with the columns which indices are in $gr_{c_0}(p)$.
2. Furthermore, we need to have a selection method:
   $$\forall p \in \{1, ...P\} : gr_1(p) = \{p\} \quad \text{and} \quad gr_0(p) = \{1, ...P\}$$
   which maps the design matrix $\mathbf{X}$ and the response variable $\mathbf{y}$ to a 0-1 vector of length P with 1 at position p if the method selects the variable p and 0 otherwise.
3. We then use the von Mises-Fisher distribution to generate replacement of the original variables by some simulations to create B new design matrices $\mathbf{X}^{(1)}, ..., \mathbf{X}^{(B)}$.
4. The SB algorithm then applies the variable selection method $select$ to each of these matrices and returns a vector of length P with the frequency of apparition of each variable. The frequency of apparition of variable $\mathbf{x}_{p\cdot}$, noted ζ_p is assumed to be an estimator of the probability of $\mathbf{x}_{p\cdot}$ being in the true model.
5. The choice of c_0 is crucial. The model might not perturbed enough (too large c_0) or the variables are chosen at random (too small c_0).

The SB algorithm (**Algorithm 1**) returns the vector $\boldsymbol{\zeta} = (\zeta_1, ..., \zeta_P)'$. Compare these values $\boldsymbol{\zeta} = (\zeta_1, ..., \zeta_P)'$ to a threshold $\zeta_{\min}$ to determine which variables are selected.

Pseudo-code

Algorithm 1 Pseudo-code for the SelectBoost algorithm

Require: $gr_{c_0}, select, B, c_0$
  $\boldsymbol{\zeta} \leftarrow \mathbf{0_P}$
  for $b = 1, ..., B$ **do**
    $\mathbf{X}^{(b)} \leftarrow \mathbf{X}$
    for $p = 1, ..., P$ **do**
      $\mathbf{x}^{(b)}_{\mathbf{p}\cdot} \leftarrow \phi^{-1}\left(\text{random-vMF}\left(\hat{\boldsymbol{\mu}}(\phi(\mathbf{X}_{\mathbf{gr_{c_0}(p)}})), \hat{\kappa}(\phi(\mathbf{X}_{\mathbf{gr_{c_0}(p)}}))\right)\right)$
    end for
    $\boldsymbol{\zeta} \leftarrow \boldsymbol{\zeta} + select(\mathbf{X}^{(b)}, \mathbf{y})$
  end for
  $\boldsymbol{\zeta} \leftarrow \boldsymbol{\zeta}/B$

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